

Human Impacts on Drainages of the Mesa Central, Mexico, and Its Genetic Effects on an Endangered Fish, *Zoogoneticus quitzeoensis*

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Abstract: *The Mesa Central of Mexico is of special conservation interest due to its high richness of freshwater fish species, of which the goodeines are one of the most representative groups. Through an integrated approach, we determined conservation priorities for goodeine populations. We based our recommendations on the genetic diversity (variation in five microsatellite DNA loci) in 10 populations of *Zoogoneticus quitzeoensis* and on an analysis of ecological (e.g., presence of exotic species), social (e.g., political situation), and environmental (e.g., pollution) information for 52 historical occurrence points for species in the genus *Zoogoneticus*. Patterns of genetic erosion and genetic diversity indices were closely associated with human impact. Recent bottleneck events were most evident in the populations from remnants of the lakes drained at the beginning of the twentieth century. We identified seven operational conservation units (OCUs), all of which should be conserved because they contain unique portions of the total variation of the species. Special attention needs to be given to increase genetic variability, recover population sizes, and reestablish contact among populations within OCUs. It is imperative to create an integrative and effective approach for the recovery and conservation of the freshwater fish diversity of Central Mexico that is based on social and natural sciences.*

Keywords: conservation units, genetic diversity, Goodeidae, human impacts, Mexican fishes

Impactos Humanos en Cuencas Hidrológicas de la Meseta Central. México, y Sus Efectos Genéticos sobre una Especie de Pez en Peligro de Extinción, *Zoogoneticus quitzeoensis*

Resumen: *La Meseta Central de México es de especial interés para la conservación debido a su alta riqueza de especies de peces dulceacuícolas, de los que los goodeinos son uno de los grupos más representativos. Por medio de un método integral, determinamos prioridades de conservación para poblaciones de goodeinos. Basamos nuestras recomendaciones en la diversidad genética (variación en el loci de cinco microsatélites de ADN) en 10 poblaciones de *Zoogoneticus quitzeoensis* y en un análisis de información ecológica (e.g., presencia de especies exóticas), social (e.g., situación política) y ambiental (e.g., contaminación) de 52 puntos de ocurrencia histórica de especies en el género *Zoogoneticus*. Los patrones de erosión genética y los índices de diversidad genética estuvieron estrechamente asociados con el impacto humano. Eventos de cuello de botella recientes fueron más evidentes en las poblaciones de los remanentes de lagos drenados a inicios del siglo veinte. Identificamos siete unidades operacionales de conservación (UOC), que deben ser conservadas porque contienen porciones únicas de la variación total de las especies. Se necesita dar atención especial para incrementar la variabilidad*

Paper submitted December 14, 2005; revised manuscript accepted June 22, 2006.

genética, recuperar los tamaños poblacionales y reestablecer contacto entre las poblaciones dentro de las UOC. Es imperativa la creación de un método integral y efectivo para la recuperación y conservación de la diversidad de peces dulceacuícolas en el centro de México que se base en información derivada de las ciencias sociales y naturales.

Palabras Clave: diversidad genética, Goodeidae, impactos humanos, peces mexicanos, unidades de conservación

Introduction

About 520 freshwater fish species are known from Mexico, of which 163 are endemic (CONABIO 1998; Miller 2005). The Mesa Central de Mexico has especially high species richness and endemism (70%) of fishes and is considered by the World Conservation Monitoring Centre as a region of special interest for the conservation of freshwater fishes (Guzmán-Arroyo 1994; Groombridge & Jenkins 1998). Freshwater environments in this region have been severely altered by anthropogenic activities. Introduction of exotic species and their parasites, pollution, water extraction, deforestation, and overfishing are just some of the causes for the decline of native species (e.g., Lyons et al. 1998; Soto-Galera et al. 1998; Domínguez-Domínguez et al. 2005a).

One of the most representative fish groups in the Mesa Central de Mexico are the goodeines, a subfamily composed of 41 species from 19 genera, most of which are endemic to the region (Doadrio & Domínguez-Domínguez 2004; Webb et al. 2004). This group presents unique reproductive characteristics, such as internal fertilization, matrotrophy, and viviparity (Parenti 1981), and is considered one of the most at-risk groups in the world (Duncan & Lockwood 2001).

The goodeine genus *Zoogoneticus* is of special concern with respect to conservation. One of the two species in the genus, *Z. quitzeoensis*, is federally listed in Mexico as endangered and the other species, *Z. tequila*, is extinct in the wild (Mexican Official Norm of Ecology-NOM-ECOL-059-2001). The populations of *Z. quitzeoensis* show high levels of genetic (mtDNA) and morphological divergence (Doadrio & Domínguez-Domínguez 2004). In addition, the species has declined greatly in distribution, and local extinctions have occurred (de la Vega-Salazar et al. 2003; Domínguez-Domínguez et al. 2005b).

There has been considerable discussion regarding how to use genetic information to identify “operational groups” for conservation. Genetic diversity is generally recognized as the base level of biodiversity (as recommended by the Convention on Biological Diversity, held in Brazil in 1992), and the exclusive use of molecular data to identify conservation units for development of conservation strategies is tempting. Nevertheless, such data should be evaluated together with historical, ecological, and social information and species distributions to obtain

a more accurate and comprehensive perspective (Crandall et al. 2000). Recently, other sources of information have been incorporated in conservation strategies, including the use of spatial distribution of genetic diversity and ecology and ecosystem functions to delimit conservation units, at the species and/or the geographic level (Moritz 2002; Luck et al. 2003; Manel et al. 2003). For the conservation of freshwater fish diversity, Doadrio et al. (1996) introduced the OCU concept, which is defined as “a continuous area limited by geographical boundaries and inhabited by one or more populations sharing the same genetic pattern.” Other factors such as conservation status and socioeconomic issues have been used to delimit OCUs (Dodson et al. 1998). We used geographic variation at microsatellite DNA loci and analyses of land use, anthropogenic disturbance, and the historical and present areas of occupancy to identify OCUs and make recommendations for the conservation of *Z. quitzeoensis*.

Methods

Samples and Data Collection

We obtained the historical locations of *Z. quitzeoensis* from the University of Michigan, Museum of Zoology; the Ichthyological Collection, Universidad Michoacana de San Nicolás de Hidalgo; and the National Collection of Fishes, Instituto de Biología, Universidad Nacional Autónoma de México. To evaluate the present distribution of *Z. quitzeoensis* and to assess environmental threats, we sampled 52 localities of historical occurrence from 2000 through 2005. The fieldwork included inspection of habitat, collection of information on the presence of exotic species, evidence of pollution, and other habitat alterations.

We used minnow traps and seine nets for sampling. All fish captured were identified and released unharmed. All field and historical records were compiled in a georeferenced database. Only localities visited at least four times were included in the study.

We preserved (in ethanol) caudal fin clips from 135 fish representing 10 populations from five biogeographic regions: Cuitzeo, Zacapu, Ameca, Middle Lerma, and the Lower Lerma (sensu Domínguez-Domínguez et al., 2006). Because of the endangered status of *Z. quitzeoensis* and

Table 1. Microsatellite diversity indices for *Zoogoneticus quitzeensis* populations sampled in central Mexico.*

Population	Biogeographic region	n	Average no. of alleles/locus	A_R	H_o	H_e	F_{IS}	$H_e W_{t(IAM/SMM/TPM)}$
El Moloya	Ameca River	10	5	4.65	0.61	0.6	-0.081	+/-/-
Magdalena	Ameca River	7	4.2	4.2	0.6	0.65	-0.018	+/+/+
Platanera	Lower Lerma	12	3.6	3.16	0.45	0.46	-0.185	+/-/+
Orandino	Lower Lerma	10	6.2	5.33	0.58	0.63	0.085	-/-/-
La Luz	Lower Lerma	9	4.6	3.18	0.45	0.49	0.163*	-/-/-
San Francisco	Middle Lerma	19	3.8	2.97	0.38	0.42	0.093	-/-/-
Zacapu	Zacapu drainage	20	3	2.93	0.51	0.4	-0.235	+/+/+
San Cristóbal	Cuitzeo drainage	12	6.2	3.67	0.5	0.46	0.145	+/+/+
La Mintzita	Cuitzeo drainage	17	4.8	3.91	0.48	0.47	-0.025	+/+/+
Belisario	Cuitzeo drainage	19	5.6	3.93	0.45	0.48	0.144	+/-/+

Key: n, sample size; A_R , mean allelic richness per locus based on the minimum sample size; H_o and H_e , observed and unbiased expected heterozygosity, respectively (Nei 1973); F_{IS} , inbreeding index ()significant departure from the Hardy-Weinberg equilibrium, $p < 0.001$; $H_e W_t$, heterozygosity excess estimated by Wilcoxon sign-rank test, based on infinite allele model (IAM), stepwise mutation model (SMM), and two-phase model (TPM); +, significant values for bottleneck detection ($p < 0.05$).

the scarcity of the species at all collection sites, sample sizes ranged from 7 to 20 individuals per population (Table 1). Individuals used for molecular analysis were from a single collection at each locality.

DNA Extraction and Microsatellite Loci Amplification

We isolated total genomic DNA following a standard cetyl-trimethylammonium bromide (CTAB) and phenol-chloroform extraction protocol (Doyle & Doyle 1987; Sambrook et al. 1989). Five microsatellites were amplified in all samples (ZT1.2, ZT1.6, ZT1.7, ZT1.9, and ZT1.43) following Boto and Doadrio (2003). The polymerase chain reaction (PCR) products were analyzed on an ABI 3700 automatic sequencer and alleles determined using GeneScan (version 3.5.1) and Genotyper (version 3.6) (Applied Biosystems, Weiterstadt, Germany).

Microsatellite Analysis

We assessed tests of Hardy-Weinberg equilibrium and linkage disequilibrium with GENEPOP (Raymond & Rousset 2000). The alpha level was corrected with the sequential Bonferroni test (Rice, 1989). Using Popgene 1.32 (Yeh & Boyle 1997), we computed for each population the number of alleles per locus (A), observed and unbiased expected heterozygosities (H_o and H_e), and the inbreeding coefficient (F_{IS} , Weir & Cockerham 1984). The observed number of alleles is highly dependent on sample size (El Mousadik & Petit 1996). To avoid this problem, we used FSTAT 2.9.3.2 (Goudet 1995) to obtain estimates of allelic richness (A_R) from 1000 resamplings of the data adjusted to the minimum sample size.

We used SPAGEDI (Hardy & Vekemans 2002) to determine the relative impact of drift (F_{ST}) versus mutation (R_{ST}) on genetic divergence. Different allele sizes at each locus were randomly permuted among allelic states (20,000 permutations) to provide a simulated distribution of R_{ST} values (pR_{ST}) with 95% confidence intervals. Un-

der the null hypothesis $R_{ST} = F_{ST}$, divergence is explained by genetic drift, whereas if $R_{ST} > F_{ST}$, stepwise mutation is more important (Hardy et al. 2003).

We used Bottleneck (Cornuet & Luikart 1996) to test for recent bottleneck events. We calculated expected average heterozygosity through simulation in the infinite allele model, the stepwise mutation model, and the two-phase mutation model. Population allele frequency distributions were also examined for mode shifts. Given the number of loci and sample sizes in our data set, we used the Wilcoxon sign-rank test to test the significance of excess heterozygosity in the three mutation models (Luikart et al. 1998).

Genetic and Field Information Plot

Indexes of environmental quality and biotic integrity were obtained from the literature for most sites (Soto-Galera et al. 1998; Lyons et al. 2000; Mercado et al. 2002). Based on the most recent national land-use/land-cover assessment (IFN 2000), we identified areas potentially damaged by human activities by using information on land use (i.e., the transformation of native vegetation to urban areas, croplands, and cultivated or induced grasslands). Associations between habitat impact and land transformation with respect to genetic diversity and bottlenecks (i.e., the number of private alleles, mean allelic richness, observed heterozygosity, and inbreeding coefficient) were visualized with ArcView (version 3.2, Environmental Systems Research Institute, Redlands, CA).

Finally, we performed a Mantel test to determine whether the percentages of land transformation and water body reduction were significantly correlated with bottleneck detection and genetic diversity, expressed as H_o and A_R . We used NTSYS version 2.1 (Rohlf 2000) and 10,000 replications. To calculate the percentage of land that was transformed, we "trimmed" the land use/land cover maps with a map representing the sub-basin system

Table 2. Ecological changes in 52 historical localities within the eight biogeographic regions in central Mexico.*

Biogeographic region	Disappeared		PL	LW	LWo	LE	LZP
	n	localities					
Ameca drainage	6	1, desiccation	2	5-0	5	4	
Armeria drainage	2	0	1	2-0	2	0	
Santiago drainage	8	3, construction 2, diversion	3	2-1	3	0	
Chapala drainage	11	2, desiccation 2, diversion	4	6-1	5	2	
Lower Lerma drainage	7	1, diversion	4	5-1	5	4	
Middle Lerma drainage	4	1, desiccation	2	2-1	2	1	
Zacapu drainage	2	0	1	2-0	2	1	
Cuitzeo drainage	12	4, desiccation 2, construction	4	5-1	4	5	
Total	52	18	21	29-5	28	17	

*Abbreviations: n, number of localities sampled with historical records of *Z. quitzeoensis*; PL, polluted localities; LW, localities with fishes; LWo, localities without fishes; LE, localities with exotic fishes; LZP, localities with *Z. quitzeoensis* populations.

of Central Mexico (sensu Domínguez-Domínguez et al., 2006) in ArcView version 3.2. In addition, we used aerial photographs 1:20000 taken by the Instituto Nacional de Estadística Geografía e Informática (INEGI) during the 2003 campaign to calculate the percentage of water surface that had been drained.

Results

Population Status

Fifty-two localities with historical information about the presence of the species were sampled between 2000 and 2005. Eighteen of these localities disappeared completely (Table 2). In the 34 remaining localities *Z. quitzeoensis* was present in only 17, indicating a local extinction of 67% (Fig. 1).

Twenty-one localities showed evidence of pollution (e.g., fetid smell due to organic decomposition, foam, or chemicals), and five of these were devoid of fish (Table 2 & Fig. 2). Fish were present at 29 of the localities sampled, and 28 of those had at least one exotic species (Table 2). The most common introduced species were carp (*Cyprinus carpio*), largemouth bass (*Micropterus salmoides*), tilapia (*Oreochromis* spp.), and several poeciliids (Table 3).

At the basin level, Santiago and Armeria basins had the greatest percentage (100%) of locally extinct populations, followed by the Chapala (80%) and Middle Lerma (75%) basins (Table 2 & Fig. 1). The Ameca basin had the fewest local population extinctions (17%; Table 2). Fewer than 10 specimens of *Z. quitzeoensis* were found in 7 of the 17 localities where the species was collected, indicating low abundances in most remnant populations. A few habitat localities, mainly springs and the upper reaches of rivers, still had large populations (i.e., localities 1, 4, 6, 8, and 9

in Fig. 1). Most of these sites were isolated and geographically distant from the closest populations.

For the 10 localities included in the microsatellite analyses, La Mintzita had the highest number of native species (15), followed by Zacapu (10) and Orandino (8), and Magdalena had the lowest number of native species (5). Three localities had 1 endemic species and Zacapu had 2. La Mintzita had the largest number of locally extinct species (3), followed by San Cristóbal (2) and Orandino, Magdalena, San Francisco, and Belisario (1). With the data obtained from the literature we determined that the only site with a low index of biotic integrity was San Cristóbal (Table 2). The historical sites where *Z. quitzeoensis* had been extirpated were generally associated with areas showing relatively severe changes in land use (Fig. 2).

Intrapopulation Diversity

Exact tests of linkage disequilibrium for each pair of microsatellite loci over all populations yielded no significant values ($p > 0.05$). Only the La Luz population showed a significant departure from Hardy-Weinberg equilibrium ($p < 0.001$, over all loci).

All loci showed high polymorphism with number of alleles per locus ranging from 11 at locus ZT1.6 to 25 at locus ZT1.7. Locus ZT1.2 showed the highest number of alleles in a single population (10 at Orandino), and ZT1.43 was monomorphic in six populations (Platanera, Orandino, San Cristóbal, La Mintzita, Zacapu, and San Francisco). Allele variation expressed as the average number of alleles per locus (A) and allele richness (A_R) was generally low (Table 1). The Orandino and San Cristóbal populations had the highest average number of alleles ($A = 6.2$), whereas Orandino and El Moloya showed the highest allelic richness ($A_R = 5.33$ and $A_R = 4.65$, respectively). The lowest allelic variation was in Zacapu Lake ($A = 3$ and $A_R = 2.93$). The observed heterozygosity values ranged from 0.38 in San Francisco to 0.61 in El Moloya (Table 1, Fig. 3). The Mantel tests showed that land transformation was significantly correlated with observed heterozygosity ($r = 0.235$; $p = 0.04$) but not with allele richness ($r = 0.089$; $p > 0.05$).

Interpopulation Diversity

All analyzed loci and 9 of the 10 sampled populations of *Z. quitzeoensis* contained unique alleles. Although Orandino and La Luz are geographically close and within the same biogeographic region, they had the highest number of private alleles (nine and eight, respectively).

Global estimates of F_{ST} and R_{ST} were high and significant ($F_{ST} = 0.2584$ and $R_{ST} = 0.4124$; $p < 0.001$). The only nonsignificant pairwise F_{ST} values ($p > 0.05$) were those for two pairs of populations that were recently isolated from each other: San Cristóbal-Belisario and El Moloya-Magdalena. All other pairwise comparisons were highly significant ($p < 0.001$), with F_{ST} values ranging

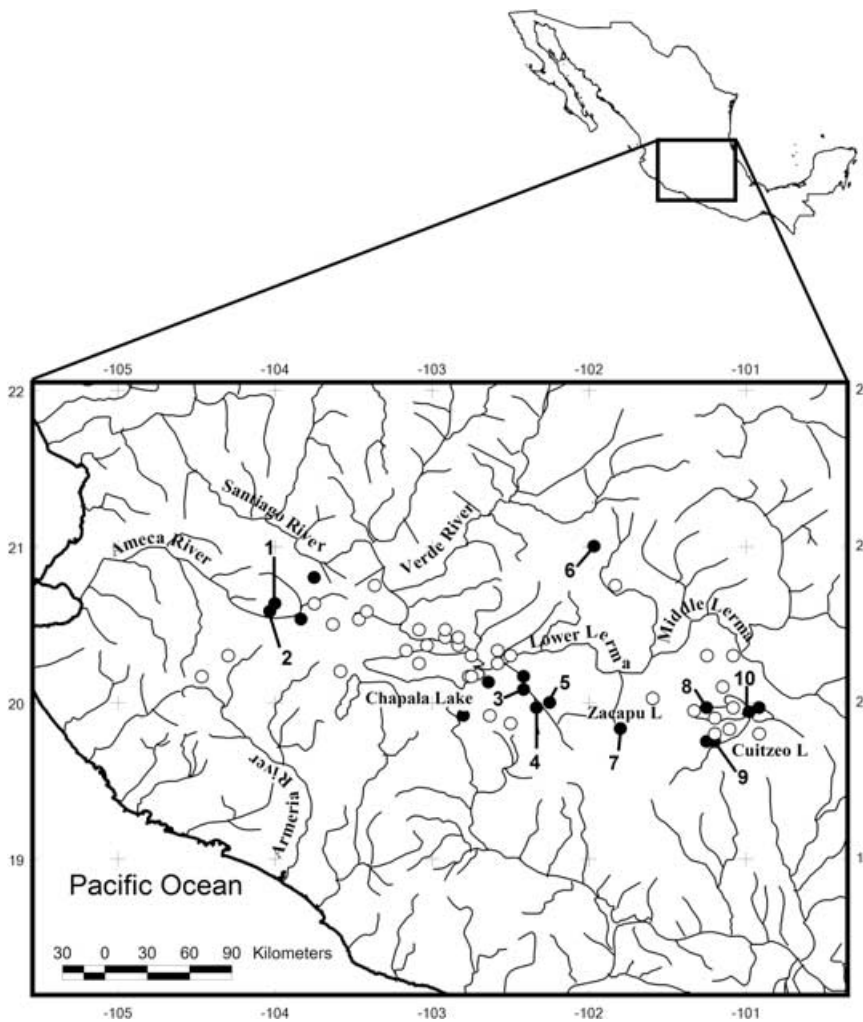


Figure 1. Local extinction in 52 populations of *Zoogoneticus quitzeoensis* in the Mesa Central, Mexico: open circles, localities where the species has not been found in the last 5 years; filled circles, localities where the species was collected in the last 5 years; numbers, localities included in microsatellite analysis: 1, El Moloya Spring; 2, Magdalena Lake; 3, La Platanera Spring; 4, Orandino Spring; 5, La Luz Spring; 6, San Francisco del Rincón Spring; 7, Zacapu Lake; 8, San Cristóbal Spring; 9, La Mintzita Spring; 10, Belisario.

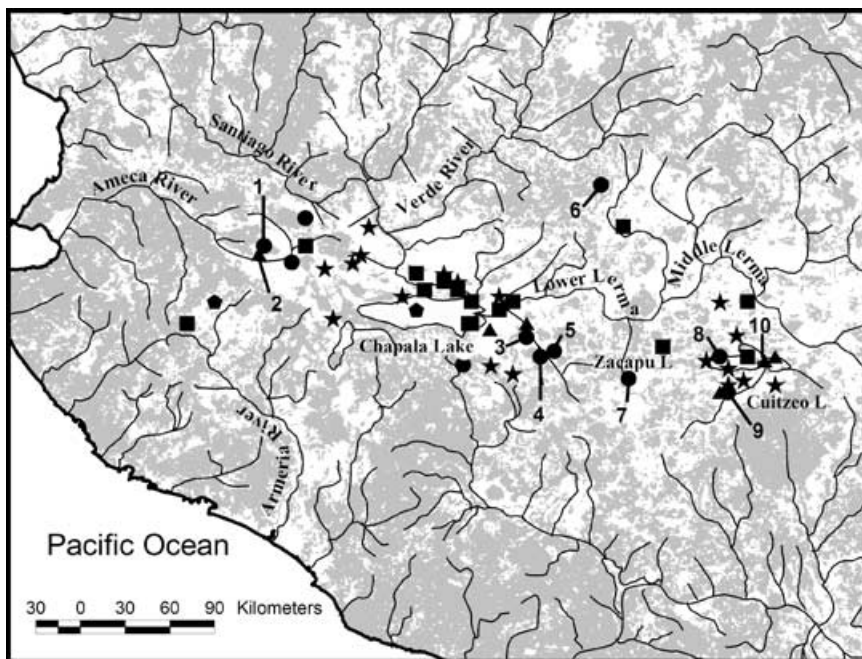


Figure 2. Land-use and land-cover assessment map and the historical localities sampled in the Mesa Central, Mexico, during field trips (2000–2005): gray, original vegetation present; white, natural vegetation totally transformed to agriculture, industry, cities, or grasslands; circles, localities with *Z. quitzeoensis*; squares, polluted localities where no *Z. quitzeoensis* were collected; triangles, polluted localities with *Z. quitzeoensis*; pentagons, no apparent perturbation, but no *Z. quitzeoensis* collected; stars, drained localities; numbers, localities included in the microsatellite analysis.

Table 3. Environmental and conservation information for the 10 populations used in the microsatellite analyses.^a

Population	Native species	Introduced species	Endemic species	EQ ^b	Extinctions	Drainage alterations	Immediate threats
El Moloja (WP)	<i>Z. quitzeensis</i> ^{c,d}	<i>Oreochromis</i> sp.	0	NI	0	after the Magdalena Lake was drained, spring remained isolated	modification of spring for recreational swimming pool; introduction of exotic species
	<i>Xenotoca eiseni</i> ^d						
	<i>X. melanosoma</i> ^d						
	<i>Ameca splendens</i> ^{c,d}						
	<i>Goodea atripinnis</i>						
Magdalena (WP)	<i>Poeciliopsis infans</i>		1	NI	1	drained in 1926 for agricultural purposes; since then lake was reduced to a few irrigation channels (Camacho 1998)	drainage for agricultural purposes; introduction of exotic species; organic pollution
	<i>Allofoca maculata</i> ^{c,d}	<i>Cyprinus carpio</i>					
	<i>Z. quitzeensis</i> ^{c,d}	<i>Ctenopharingodon idella</i>					
	<i>G. atripinnis</i>	<i>Oreochromis</i> sp.					
	<i>X. melanosoma</i> ^d						
Platanera (WP)	<i>X. eiseni</i> ^{d,f}		0	NI	0	reduction of migration rates due to isolation of the spring induced by highly polluted waters and diversion of natural water flow for agricultural purposes (Medina-Nava 2003); 17% of spring drained for agricultural purposes and water extraction	increase of the pollution in the Lerma River (Soto-Galera et al. 1998); introduction of exotic species
	<i>Allophoborus robustus</i> ^d	<i>C. carpio</i>					
	<i>Z. quitzeensis</i> ^{c,d}	<i>C. idella</i>					
	<i>Xenotoca variata</i>	<i>Oreochromis</i> sp.					
	<i>G. atripinnis</i>	<i>Micropterus salmoides</i>					
Orandino (WP)	<i>Skiffia multipunctata</i> ^d	<i>Poecilia reticulata</i>					
	<i>P. infans</i>						
	<i>Chapalichthys encaustus</i> ^d						
	<i>Chirostoma jordani</i>						
	<i>S. multipunctata</i> ^d	<i>C. carpio</i>	1	medium ***	1	reduction of migration rates due to isolation of the spring induced by highly polluted waters and diversion of natural water flow for agricultural purposes (Medina-Nava 2003)	polluted environments persist in area; introduction of exotic species
	<i>C. encaustus</i> ^d	<i>C. idella</i>					
	<i>Z. quitzeensis</i> ^{c,d}	<i>Xiphophorus bellerii</i>					

continued

Table 3. (continued)

Population	Native species	Introduced species	Endemic species	EQ ^b	Extinctions	Drainage alterations	Immediate threats
La Luz (WP)	<i>G. atripinnis</i>	<i>Poecilia spbenopbs</i>					
	<i>P. infans</i>						
	<i>Algansea tincella</i>						
	<i>Notropis josealvarezii</i> ^{e,f}						
	<i>A. robustus</i> ^d						
	<i>S. multipunctata</i> ^d		0	medium ***	0	reduction of migration rates due to isolation of spring induced by highly polluted waters and diversion of natural water flow for agricultural purposes (Medina-Nava 2003)	polluted environments persist in area; introduction of exotic species; extraction of water for human and irrigation supply
	<i>C. encarnatus</i> ^d						
	<i>Z. quitzeoensis</i> ^{c,d}						
	<i>G. atripinnis</i>						
	<i>P. infans</i>						
	<i>Lampetra geminis</i> ^c						
	<i>A. robustus</i> ^d						
	<i>Z. quitzeoensis</i> ^{c,d}		0	medium ***	1	highly polluted environments around spring (Soto-Galera 1998) reduced migration rates between contiguous water bodies	pollution persists in surrounding area; introduction of exotic species; water use for recreational and human supply
	<i>X. variata</i>						
	<i>G. atripinnis</i>						
Zacapu (EP)	<i>A. tincella</i>						
	<i>Notropis callientis</i> ^f						
	<i>P. infans</i>						
	<i>A. robustus</i> ^d						
	<i>Z. quitzeoensis</i> ^{c,d}		2	medium ***	0	drainage of original marshy area (15,000 ha) in 1907 for agricultural purposes; lake now only 32 ha (Guzmán 1985)	introduction of exotic species; occasionally polluted by wastewaters
	<i>Allotoca zacapuensis</i> ^{d,e}						
	<i>G. atripinnis</i>						
	<i>X. variata</i>						
	<i>Hubbsina turneri</i> ^{c,d,e}						
	<i>A. robustus</i> ^d						
	<i>Skiffia lermiae</i> ^{c,d}						
	<i>Chirostoma bumboldtianum</i>						
	<i>A. tincella</i>						
	<i>N. callientis</i>						
	<i>Z. quitzeoensis</i> ^{c,d}		0	medium ***	2	isolation due to total drainage of west part of Cuitzeo Lake (Tamayo & West 1964); situation persists today	use of the spring water for human supply; introduction of exotic species

continued

Table 3. (continued)

Population	Native species	Introduced species	Endemic species	EQ ^b	Extinctions	Drainage alterations	Immediate threats
La Mintzita (EP)	<i>G. atripinnis</i>	<i>Xiphophorus maculatus</i>					
	<i>S. lernae</i> ^{c,d,f}	<i>X. bellerti</i>					
	<i>A. robustus</i> ^d						
	<i>Xenotoca variata</i> ^d						
	<i>P. infans</i>						
	<i>C. jordani</i> ^f						
	<i>Z. quitzeoensis</i> ^{c,d}	<i>C. carpio</i>	0	medium**	3	70% loss due to agricultural expansion and continuous extraction of ground water, which reduced 25% of water flow of spring (Guzmán 1985)	reduction of spring flow due to ground water extraction and its use for human supply; introduction of exotic species
	<i>A. robustus</i> ^d	<i>C. idella</i>					
	<i>Allotoca catarinae</i> ^d	<i>M. salmoides</i>					
	<i>Allotoca dugesi</i> ^{c,d,f}	<i>Oreochromis</i> sp.					
	<i>G. atripinnis</i>	<i>P. reticulata</i>					
	<i>X. variata</i> ^d						
	<i>S. lernae</i> ^{c,d}						
Belisario (EP)	<i>H. turneri</i> ^{c,d,f}						
	<i>N. callentis</i>						
	<i>Notropis sallei</i> ^f						
	<i>Scartomyzon austrinus</i>						
	<i>C. humboldtianum</i>						
	<i>Chirostoma charari</i> ^c						
	<i>A. tincella</i>						
	<i>P. infans</i>						
	<i>Z. quitzeoensis</i> ^{c,d}	<i>Oreochromis</i> sp.	1	poor*	1	complete drainage of the Cuitzeo Lake in winter 1941 when construction of Cointzio Dam was finished (De Buen 1943)	introduction of exotic species; water pollution; drainage
	<i>A. dugesi</i> ^{c,d}	<i>M. salmoides</i>					
	<i>Neotoca bilineata</i> ^{c,d}	<i>C. carpio</i>					
	<i>C. peraticus</i> ^d						
	<i>H. turneri</i> ^{c,d,f}						
	<i>C. jordani</i>						

^a Abbreviations: EQ, index of environmental quality; NI, no information; WP, populations that correspond to the west group; EP, populations that correspond to the east group.

^b Citations: * Mercado et al. 2002; ** Lyons et al. 2000; *** Soto-Galera et al. 1998.

^c Species with status (NOM-ECOL-059-2001).

^d Includes the information from Domínguez-Domínguez et al. (2005b).

^e Endemic species.

^f Species locally extinct.

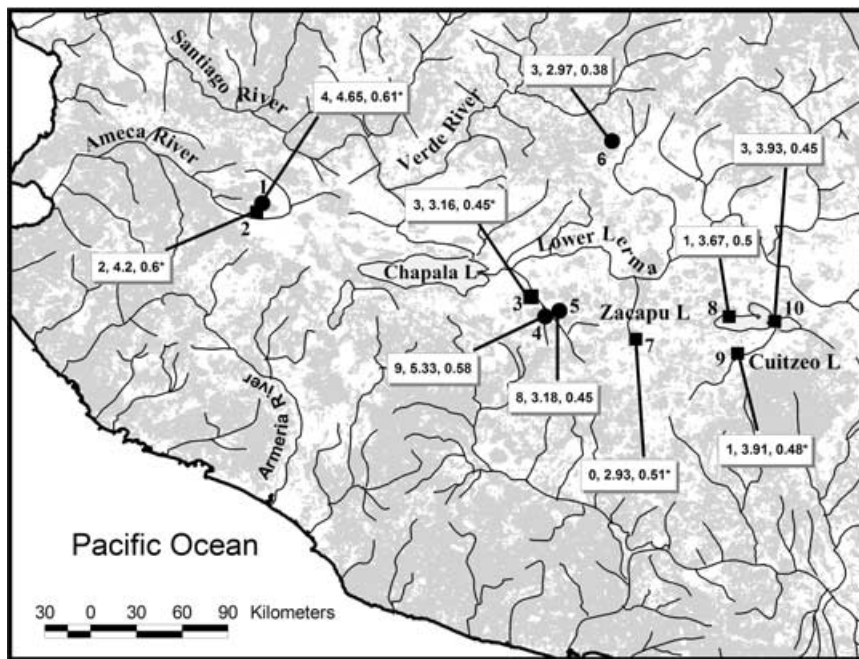


Figure 3. Land-use and land-cover assessment map of the Mesa Central Mexico: gray, original vegetation present; white, natural vegetation totally transformed for agriculture, industry, cities, or grasslands; squares, populations with significant bottlenecks; circles, no significant bottlenecks; numbers in boxes, numbers of private alleles, mean allele richness, and observed heterozygosity, respectively; *, populations with negative F_{IS} values; numbers, localities included in microsatellite analysis.

from 0.046 for La Mintzita-San Cristóbal to values as high as 0.455 for La Luz-San Francisco.

Based on the allele size permutation test, R_{ST} values were significantly greater than F_{ST} values in most of the cases where comparisons were made between populations west of the Lower Lerma River (WP, western populations) and those east of the Zacapu and the Middle Lerma River (EP, eastern populations). In these instances stepwise mutation had a greater effect on genetic differentiation than drift (i.e., F_{ST} underestimated the actual levels of genetic structure; thus, R_{ST} would be a better estimator). The highest pairwise R_{ST} values were close to the upper theoretical boundary for comparisons of La Platanera with San Cristóbal, La Mintzita, and Zacapu ($R_{ST} = 0.855\text{--}0.882$).

Population Bottleneck

The tests for population bottleneck supported previously documented declines in most of the populations of *Z. quitzeoensis*. Seven of the studied populations showed significant values of heterozygosity excess ($p < 0.05$) for at least two of the three mutation models tested (Table 1). Four of the populations (Magdalena, Zacapu, San Cristóbal, and La Mintzita) had significant values for all three models. Although the qualitative mode-shift model cannot be used with a high confidence for samples with fewer than 30 individuals (Cornuet & Luikart 1996), in our analyses samples from Magdalena, Platanera, Zacapu, San Cristóbal, and La Mintzita showed allele frequency shifts indicative of recent population declines. The results are in most cases comparable with those obtained for the Wilcoxon test. Only Belisario, which showed significant heterozygosity excess values for infinite allele model and

two-phase mutation model, showed a normal L-shaped distribution as would be expected for a stable population in mutation-drift equilibrium (Luikart et al. 1998). The Mantel test showed a significant correlation between percentage of water body drainage and significant bottleneck detection ($r = 0.372$; $p = 0.016$).

Discussion

This study is one of the first to apply extensive fieldwork, information on habitat degradation, geographic information systems (GIS), and genetics to the conservation of fishes, particularly to the endangered Mexican ichthyofauna. The OCUs (Doadrio et al. 1996) were determined based on information from previously described biogeographic regions (Domínguez-Domínguez et al. 2006), genetic, environmental, and ecological data. We considered the OCUs the most appropriate criterion for the identification of management units for freshwater fish species (Doadrio et al. 1996): those that have limited dispersal capability and are circumscribed to territories with well-defined boundaries such as rivers or lakes.

Environmental and Genetic Information

Across the distribution of *Zoogoneticus quitzeoensis* we found a high percentage of local population extinction and a dramatic reduction in population sizes, as well as evidence of fragmentation. One of the most common human alterations in the area is the drainage of water bodies. Evidence for this is provided by the absence of water in 18 localities we studied. The devastation of water bodies

of Central Mexico was induced for agricultural purposes at the beginning of the twentieth century (Table 3) and caused the direct drainage of lakes, lower water inflow due to the diversion of rivers, and reduced water tables. The most affected areas are the La Mintzita, Zacapu, Chapala, Magdalena, and Cuitzeo lakes, which in some cases have lost up to 90% of their surface area (e.g., Zacapu Lake). Other important alterations to the water system of the Mesa Central include the government plan for the introduction of exotic fish species as a food supply or as biological control (Tables 2 & 3). This has had a negative impact on the native species, leading in some cases to their extinction (Zambrano et al. 1999; Domínguez-Domínguez et al. 2005b).

All these situations are reflected in the estimates of genetic diversity, the values of which were similar or lower than those found for other endangered freshwater fishes (Salgueiro et al. 2003; Mesquita et al. 2005; Lage & Kornfield 2006), and the evidence of significant bottlenecks in all populations studied from these areas, with the exception of El Moloya (Table 1). Additional evidence of the impact of humans on the genetic structure of the species includes the significant F_{ST} values but nonsignificant F_{ST} versus R_{ST} values among populations from the Cuitzeo basin (La Mintzita and Belisario), which are only 34 km apart. The same situation was observed between La Luz and Orandino, which are 4 km apart. This indicates that genetic drift due to relatively recent fragmentation is more important than ancient isolation and mutation in explaining genetic divergence between these two pairs of populations.

The pollution caused by industry, wastewater from urban areas, and changes in land use (Fig. 2) is also reducing and fragmenting habitat in the water bodies of Central Mexico. Pollution is most evident in the Santiago, Lower and Middle Lerma river basins, and the Lake Chapala Basin, where there was a large reduction in occurrence points (Table 2 & Fig. 1). This damage has been documented in other studies of fish diversity in the region (Soto-Galera 1998; Soto-Galera et al. 1999; Lyons et al. 2000; Mercado-Silva 2002). The populations with the lowest values of genetic diversity were located in these highly polluted basins (Table 1). Populations at La Platanera (which had the highest number of introduced species) and San Francisco showed the lowest overall genetic diversity ($A_R = 3.16$, $H_o = 0.45$, and $A_R = 2.97$, $H_o = 0.38$, respectively).

Most of the surviving wild populations are restricted to springs or water bodies fed by springs. Pollution levels are lower in these areas than in rivers because the spring flows maintain a good water quality. The populations in some of these springs, such as those at El Moloya, Orandino, and La Mintzita, are the most genetically diverse of those surveyed. Water bodies that have been historically stable in terms of water quality and size (e.g., Orandino and San Francisco) showed no evidence of re-

duced population size (Table 1 & Fig. 3). It appears that the size of the water body (which affects population size) is important in maintaining genetic diversity. For example, in the Lower Lerma Basin, Orandino (approximately 5.2 ha) had the highest diversity ($A = 6.2$, $A_R = 5.7$, and $H_o = 0.58$), whereas La Luz and Platanera, which are almost three times smaller (approximately 1.7 ha and 2.1 ha), had diversity estimates around 30–40% lower ($A = 4.6$, $A_R = 3.18$, and $H_o = 0.45$ and $A = 3.6$, $A_R = 3.16$, and $H_o = 0.45$, respectively). The importance of water bodies fed by springs is corroborated by the high richness of native species found within them (i.e., La Mintzita, Orandino, and Zacapu) (Table 3).

Although springs are important refuges for many species, conservation strategies cannot focus only in these areas. Their geographical isolation and the lack of gene flow among some of them combine to increase the possibility of genetic degradation (e.g., La Luz and San Cristobal, Table 1).

Genetic Structure

The results of the allele-size permutation test demonstrated that stepwise mutation had a greater effect than drift on genetic divergence ($R_{ST} > F_{ST}$) between western (WP) and eastern (EP) populations of *Z. quitzeoensis*. This indicates that these two groups of populations have been isolated long enough for mutations to accumulate (Hardy et al. 2003). This corroborates the finding that sequence divergence in the mitochondrial cytochrome *b* gene was higher between EP and WP populations of *Z. quitzeoensis* (2.5–3%) than in any other intraspecific comparison of goodeines (Doadrio & Domínguez-Domínguez 2004).

For comparisons between populations belonging to EP and WP, values of R_{ST} ranged from 0.535 for Magdalena-San Francisco to 0.882 for La Platanera-Zacapu. The main exception was a western population (La Luz) for which R_{ST} was not significantly greater than F_{ST} in any comparisons with populations from EP, and its lowest F_{ST} value (0.271) was obtained when it was compared with Orandino; they are in the same basin. All other comparisons involving La Luz, either with EP or WP, showed similar values of genetic differentiation ($F_{ST} = 0.345$ – 0.455). La Luz was the only population with a significant departure from Hardy-Weinberg equilibrium and no evidence of a recent bottleneck event, but it did have a high inbreeding index (Table 1). This, together with the high number of private alleles in this population (the highest after the Orandino population) and the significant F_{ST} values but nonsignificant F_{ST} versus R_{ST} when compared with the populations from Platanera, Orandino, and La Luz, indicates that genetic drift in the Lower Lerma Basin has been more intense than in other areas occupied by *Z. quitzeoensis*.

The current population structure within each of the two groups (WP and EP) of *Z. quitzeoensis* could be explained by two alternative hypotheses: (1) the migration rate among populations has been larger than the mutation rate, so new mutational variants spread over all populations (e.g., Moloya-Magdalena, Belisario-San Cristóbal) or (2) human activities have caused relatively recent fragmentation and reduction of populations. The first explanation is counterindicated by the high F_{ST} values among populations and by the high rate of population loss in the area. The second hypothesis is supported by our environmental data and genetic results. The results of the allele size permutation test indicated that genetic drift explains the divergence among populations within both WP and EP.

OCU Identification

Because the conservation of freshwater fishes in Central Mexico needs to be an integrative task, we identified which populations are in greater need of protection taking into consideration biogeographic, genetic, environmental, social, and political issues. From the 10 populations studied, we identified seven OCUs: (1) Cuitzeo, including the Belisario, La Mintzita, and San Cristóbal populations; (2) Magdalena, including the Magdalena and El Moloya populations; (3) Zacapu; (4) San Francisco del Rincón; (5) Orandino; (6) La Luz; and (7) La Platanera, with OCUs 3–7 corresponding to single populations. All OCUs represent separate biogeographical regions, except La Luz, Orandino, and La Platanera, which belong to the same biogeographic region.

El Moloya and Magdalena within the Ameca drainage were not divergent based on F_{ST} and shared most of their alleles. In such cases environmental and social aspects should be taken into consideration when conservation strategies are proposed. El Moloya and Magdalena show high values of A_R and H_o (Table 1), but the former has more private alleles (4) than the latter (2). The environmental quality of El Moloya is higher because it is a spring with a high water flow. The endangered restricted species *Ameca splendens* occurs at this locality. In contrast, the Magdalena population suffers from intense human pressure because of changes in land use and the diversion of water for irrigation, but it still contains the endemic species *Allotoca maculata*.

In Cuitzeo basin the La Mintzita population showed low but significant F_{ST} values compared to Belisario and San Cristóbal. La Mintzita and Belisario are small lakes fed by large spring flows, but La Mintzita had a greater number of native species and is classified as a regional natural park. In contrast, Belisario suffers from high human pressure, high fluctuations in hydrologic regime, and pollution.

The three populations from Lower Lerma (Orandino, Platanera, and La Luz) differed in levels of genetic diversity and inbreeding and had high numbers of private al-

leles. This, together with the high environmental quality of these water bodies suggests that they should be considered different OCUs in the Lower Lerma Basin.

For OCUs corresponding to single populations, most showed high values of F_{ST} when compared with other populations, and most of these OCUs had private alleles. Zacapu Lake was the only population with low F_{ST} values when compared with populations from the Cuitzeo basin. In addition, no private alleles were detected, and it had the lowest average of alleles per locus and A_R . The classification of Zacapu Lake as a separate OCU was based on its high environmental quality, high number of native and endemic species, and its designation as a regional natural park.

Conservation and Management Implications

The devastation of the basins of Central Mexico apparently has affected the genetic variability and structure of *Z. quitzeoensis*. This is exemplified by the bottlenecks detected in the populations inhabiting the lakes drained at the beginning of the twentieth century, such as Zacapu, La Mintzita, Magdalena, and the two Cuitzeo Lake populations (San Cristóbal and Belisario). The probability of population extinction is heightened by increased demographic stochasticity, which causes inbreeding, loss of genetic variation, and fixation of deleterious alleles (Saccheri et al. 1998). Habitat reclamation and stability should be the most important goal of conservation strategies if genetic degradation in depressed populations (following a bottleneck, such as in Zacapu and La Mintzita) is to be reversed. In populations such as San Cristóbal and Magdalena, where apparently the bottleneck was not as dramatic as in other populations, reestablishment of natural conditions, recovery of population numbers, and maintenance of genetic diversity needs to be taken into consideration.

Pollution and introduction of exotic species have forced some populations of native species to live in refuges (i.e., La Luz and San Francisco del Rincón), which has led to reductions in population size and genetic variability. Although high genetic diversity indices are found in small populations such as Orandino, special attention still needs to be given to genetic and demographic characteristics if conservation and management strategies are to be successful. The degradation of habitat should be stopped and habitat conservation is the most important conservation tool.

The fact that each OCU identified here contains unique portions of the total variation of the species suggests that they all should be conserved. In addition, we conclude that gene flow among OCUs is restricted or does not exist. This indicates there is little opportunity for reestablishment of populations and their particular genetic characteristics after local extinction. The reestablishment of

contact between populations, where historical contact can be demonstrated (e.g., El Moloya and Magdalena), is a key factor for the long-term conservation of *Z. quitzeoensis*, even more so when one considers the increasing effects of human activities on its natural range.

None of the natural protected areas of Central Mexico include populations of *Z. quitzeoensis*, and, except for Cuitzeo and Chapala, none of the hydrologic priority regions from Central Mexico include the species. Therefore, other sources of information should be considered in recognizing important areas for the conservation of the freshwater diversity of Central Mexico.

The success, or lack thereof, of translocation programs and their effect on the reversal of genetic degradation in wild populations is an ongoing debate (Tallmon et al. 2004; Geist & Kuehn 2005). Considering the high levels of genetic divergence we found, translocations between different OCUs should be avoided except as a last resort. Other approaches to management and recovery need to be implemented in response to the high degree of degradation of the aquatic ecosystems in Central Mexico and its impact on the population dynamics and genetic diversity. We recommend the development of captive breeding programs and semiartificial breeding and perhaps translocations of populations with previous contact as a last resort.

Integration of data on the distribution of genetic diversity with historical or current environmental and ecological data should play a role in any future conservation plans for *Z. quitzeoensis*. It is imperative to create an integrative approach involving social and natural sciences for the recovery and conservation of such highly fragmented populations. The survival of this species depends critically on minimizing or reversing the habitat degradation that has occurred.

Acknowledgments

We thank the following people for their support during the fieldwork: I. Dibble, J. Nightingale, J. Moree, J. M. Artigaz, L. Couvreur, K. de Jong, R. Pérez, L. Escalera, A. Molina, K. Arias, M. Simón, A. Pompa, H. Mejía, and R. Rosas. L. Alcaraz assisted in the laboratory work. S. Young and P. Bloor reviewed the English text and the anonymous referees helped improve the manuscript. This study was partially supported by the Hobbyist Aqualab Conservation Project, which receives funds from European and North American aquarist associations, and the Chester Zoo in England. O.D.D. thanks the Consejo Nacional de Ciencia y Tecnología (CONACYT) for providing a scholarship. Fieldwork was partially funded by the program PAPIIT-UNAM-IN220605 and CONACYT (grant number 47233) to G.P.P.D.L.

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